

The end of all things

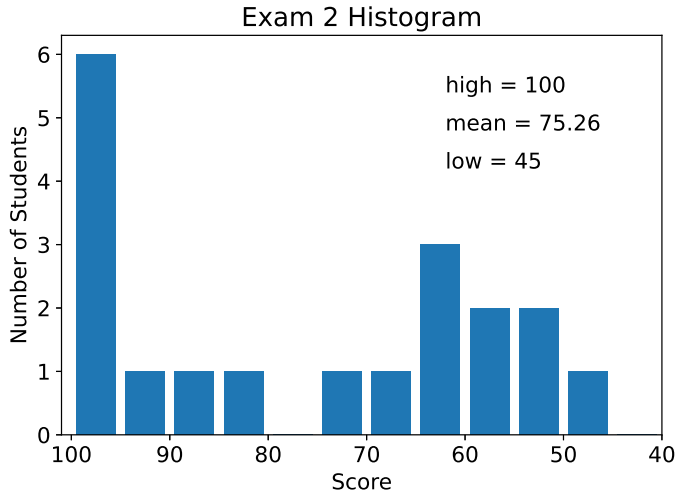
EGR 445/545

GVSU

Plan for today

- exam 2 review
- Jacobians in other frames
 - three link planar Jacobian discussion
- Jacobian questions
- Project 2 demos

Exam 2 Histogram



Jacobians in Other Frames

Jacobians in Other Frames

- why?
 - sometimes the physical meaning of the singularity is clearer in other frames
 - often frame 1 is the most helpful
- how?
 - rotate v_{tip} and ω_N into the desired frame before taking the partial derivatives

Singularity Note

The determinant being zero is one way to find when a matrix cannot be inverted. Another way of saying the same thing is that the matrix must be *full rank*:

- all of its rows must be linearly independent of one another
- all of its columns must be linearly independent of one another